

Muhammad Saad Ali

+92-320-000-6080 | [Email](#) | [Linkedin](#) | [Github](#)

RESEARCH INTEREST

My research focuses on building intelligent systems that are usable, trustworthy, and deployable in resource-constrained real-world environments. I am particularly interested in human-centered AI, HCI-driven system design, and the integration of IoT and contextual sensing with AI models to support decision-making. I aim to develop frameworks and systems that reduce cognitive load, improve interpretability, and align system behavior with human workflows, with a broader goal of advancing dependable, accessible, and equitable technology for underserved communities.

PUBLICATIONS

C=CONFERENCE, P=POSTER

- [P.1] Ali, M.S., Khan, D.U., Intazar, L.A., *et al.* (2024). **Poster: Bridging IoT Gaps in Developing Regions with LLMs**
- [C.1] Ali, M.S., Khan, Ahmed, L.I., Khan, D.U., *et al.* (2025). **Kissan-Dost: Bridging the Last Mile in Smallholder Precision Agriculture with Conversational IoT.** *Under review (PerCom).*

EDUCATION

Lahore University of Management Sciences (LUMS)

B.S. in Computer Science

Aug 2021 – July 2025

PROJECTS

Kissan Dost — LLM + IoT Precision Agriculture

Jan 2024 – Present

- Researched and developed an end-to-end AI + IoT system for precision agriculture, integrating low-cost sensor networks with LLM-based decision support for farmers.
- Engineered custom ESP32-based hardware with solar-powered duty cycling and long-range communication for real-time soil data collection across 1000+ acres.
- Built a RAG pipeline linking sensor data with verified agronomic knowledge bases, enabling natural language recommendations, disease identification, and early warnings via WhatsApp interface.
- Conducted a longitudinal user study over a full crop cycle to evaluate adoption and utility, observing usage patterns in irrigation management, disease diagnosis, and weather preparedness.
- Implemented human-in-the-loop validation with expert farmers to assess LLM recommendation accuracy and contextual reliability.

Velocity — Reconciliation Management System for Accelerant

July 2025 – Present

- Intervened in an outsourced manual reconciliation workflow to identify inefficiencies and formulate the research problem, leading to the development of an AI-enabled system to automate high-volume financial transaction matching and reporting.
- Designed LLM-based pipelines for discrepancy detection, policy matching, premium reconciliation, and structured reporting, enabling accurate processing of non-standard financial data across multiple MGAs.
- Architected the system to ensure GDPR, SOC 2, and ISO compliance, embedding privacy, security, and auditability into all stages of the data pipeline.
- Deployed the system with MGA “Triad,” processing over 100K transactions and reducing reconciliation time from 4 to 5 days to under 15 minutes per cycle, with 98%+ matching accuracy and an 80% reduction in manual review effort.

Submissions - Document parser for Accelerant

July 2025 – Present

- Identified inefficiencies in broker submission handling and designed an automated processing system to standardize high-volume insurance documents and reduce manual underwriting effort.
- Developed LLM-powered parsing pipelines using LlamaParse and DeepInfra models to transform heterogeneous broker PDFs into structured, underwriting-ready data.
- Implemented a resilient architecture with asynchronous job handling, history tracking, and error recovery to ensure high throughput, reliability, and SOC 2 / GDPR / ISO compliance.

- Deployed the system at scale, achieving near-instant processing times per submission and enabling consistent, low-latency data flows across multiple brokers and document types.

Voicera AI — Voice AI Agent for Salons & Hairdressers

July 2025 – Present

- Led the design and deployment of a voice-based AI receptionist, integrating LLMs with scheduling logic to automate booking and customer interactions in real-world salon environments.
- Conducted structured evaluations across partner salons in the UK, demonstrating major gains in responsiveness and reduced booking latency, validating its usability in operational workflows.

Rehash Cloud — AI Agent Marketplace & CRM Automation

2025

- Developed an AI-powered CRM automation platform and workflow builder, architecting robust agent pipelines for high-volume lead engagement.
- Delivered the MVP from concept to deployment, integrating communication and billing APIs to support scalable, production-ready automation for enterprise clients.

Credit Scoring Engine for Pakistan

Jan 2024 – May 2024

- Developed ML models (TabNet) to generate credit scores and credit limits for microfinance banks.
- Engineered feature pipelines using KYC and transaction data (e.g., shop size, sales, location), improving scoring accuracy and interpretability for risk assessment.

Multi-Card OCR

Jan 2024 – Mar 2024

- Built OCR models to digitize Pakistan's NIC and fuel cards (Shell, Total Parco, Hascol), enabling automated document processing.
- Designed detection and recognition pipelines using YOLOv7/v8, improving data extraction reliability for onboarding workflows.

Thenga — E-commerce Website

May 2023 – Aug 2023
[Website](#)

- Designed and launched a complete e-commerce platform using Bubble for local service providers.

EXPERIENCE

Apollo Studios

Jan 2025 – Present

AI Software Engineer

- Built AI systems for underwriting and reconciliation at [Accelerant](#), working closely with cross-functional teams to design, build, and deploy LLM-driven data processing pipelines with secure, compliant architectures.
- Led the development of MVPs for early-stage startups at [Hypergrowth](#) (venture building wing of Apollo), leading technical development from concept to deployment and enabling successful fundraising outcomes.

Muawin

Jan 2024 – Jan 2025

Junior Software Engineer

- Contributed to the development of Muawin's credit scoring engine and administrative platform, focusing on data-driven modeling and system reliability.

LUMS

Fall 2023 – May 2025

Research Assistant & Teaching Assistant

- **Research Assistant, SYSNET labs** (Supervisors: [H. Alizai](#), [N.A. Bhatti](#))
Contributed to the development and deployment of KissanDost, an AI+IoT precision agriculture system across large-scale farms.
- **Teaching Assistant, Introduction to Programming** (Instructor: [Dr. Awais](#))
Taught foundational programming concepts through labs and tutorials, supported assessments, and guided 200+ students in developing core computational thinking skills.
- **Teaching Assistant, Linear Algebra** (Instructor: [Dr. Imran Anwar](#))
Assisted 250+ students in applying linear algebra concepts to computational and machine learning problems through weekly tutorials and problem-solving guidance.

SKILLS

- **Programming Languages:** C++, C, Python, Haskell, Solidity, JavaScript, TypeScript
- **Web & App Development:** HTML, CSS, Django, React, Flutter, React Native, Bubble, Webflow
- **Database Systems:** MySQL
- **Data Science & Machine Learning:** NumPy, Pandas, Matplotlib, scikit-learn, OpenCV, TensorFlow
- **Cloud & DevOps:** Docker, AWS, Kubernetes, Google Cloud Run
- **DevOps & Version Control:** Git, Bash, WebPageTest, WebPageReplay
- **Other Tools & Technologies:** LaTeX